

# Software Requirement Specification

## Project : Fleet Management

### Team 28 [ Fleetix ]

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## Problem Statement

Open source route optimization using Google APIs and other optimization techniques for employee transportation, specifically addressing scenarios involving multiple pickups and drop-offs.

## System Requirements

The core requirement involves efficiently transporting employees with multiple pickup and drop-off locations. This can be categorized into two primary scenarios:

- **Many-to-One** : Multiple employees are picked up from different locations and transported to a single destination (e.g., office or event).
- **Many-to-Many** : Special case of the first scenario, where employees are picked up from different locations and dropped off at various destinations, possibly across various times.

For implementing these, following are the smaller specific features, the client has requested :

- **Create a Road Network Graph** - Represent the road network for the city/town in a graph data structure. This graph will be used for route calculations.
- **Geocode Employee Locations** - Convert employee addresses to geographic coordinates (latitude and longitude) for mapping onto the road network.
- **Constraints** : Such as vehicle capacity, time windows, and maximum route duration should be considered in final routing.

Apart from this main problem, following are some suggested features (or additional considerations) from the client :

- **User Interface** : Develop a user-friendly interface for inputting employee data, visualizing routes, and managing transportation schedules.
- **Dynamic Rerouting** : Implement dynamic rerouting capabilities to adapt to unexpected events like traffic congestion or road closures.
- **Real-time Traffic** : Integrate real-time traffic data from services like Google Maps or TomTom to improve route accuracy.

The client has an ongoing project, which he would later integrate this code with. Thus, would prefer a tech stack similar to it. The following are the current product's tech stack :

- Frontend is in next (and typescript)
- Backend is in next.js
- Database is PostgreSQL

## User Profiles

### 1. System Administrators

- Add / Remove / Edit drivers and employees
- Update constraints like vehicle capacity, routes, wait time and maximum acceptable delays

### 2. Drivers

- View routes (past and scheduled)
- Passenger details like pick-up location, drop-off location and contact details

### 3. Employees

- View routes (past and scheduled) along with co-passengers, driver and vehicle details
- Pickup/drop-off (location and time) preference settings

## Use Cases

NOTE: Since the use case of the given problem statement is very specific with not many use cases for the users, we would consider any functionality doing an isolated task as a use case. While this deviates from the definition of "relation between actors", it keeps up with the goal of breaking down the problem for better understanding and implementation.

| S. No. | Use Case Name                                   | Description                                                                 | Release |
|--------|-------------------------------------------------|-----------------------------------------------------------------------------|---------|
| 1      | Geocoding                                       | Extracting (longitude, latitude) out of an address string using Google APIs | R1      |
| 2      | Road Network                                    | Develop / Extract (from a library) a detailed road network graph            | R1      |
| 3      | Employee Pick-up / Drop-off Location Preference | Option for the employees to set their preferred pick-up / drop-off location | R1      |
| 4      | Routing (Many-to-One)                           | Choosing and implementing an optimal algorithm for routing (a               | R1      |

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|   |                          | singular vehicle) through the extracted longitudes and latitudes                                               |    |
| 5 | Visualise Route          | Appropriately visualise the routes with detailing for driver and employees                                     | R1 |
| 6 | Adding Constraints       | Fine tune the algorithm to work with predefined constraints like vehicle capacity, maximum delay and wait time | R2 |
| 7 | Ride History and Reports | Employees and admins can view past rides, including routes taken and trip details.                             | R2 |
| 8 | Many-to-Many Routing     | Multiple vehicle routing                                                                                       | R2 |
| 9 | User Interface           | Develop interactive and user-friendly web-based applications for driver, employees and system administrator    | R2 |

## Use Case Description

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| UseCase Number | UC-001                                                                                                                                                                                                                                                       |
| Use Case Name  | Geocoding                                                                                                                                                                                                                                                    |
| Overview       | Extracting (longitude, latitude) out of an address string using Google APIs                                                                                                                                                                                  |
| Actors         | System                                                                                                                                                                                                                                                       |
| Pre-Condition  | Employee addresses are available for processing.                                                                                                                                                                                                             |
| Flow           | <p>Main Flow</p> <ol style="list-style-type: none"> <li>1. The system receives an address string from the input dataset.</li> <li>2. The Google Geocoding API processes the address and returns corresponding latitude and longitude coordinates.</li> </ol> |
|                | <p>Alternate Flow</p> <p>2a. If the address could not be decoded by the Google API, appropriate errors are used. And system admins / employees would be prompted to reword the address string.</p> <p>*a. Any errors, at any stage, should prompt the</p>    |

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|                | admin / employee to reword the address string.                     |
| Post-Condition | Successfully geocoded locations are returned for route processing. |

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| UseCase Number | UC-002                                                                                                                                                                                                                  |
| Use Case Name  | Road Network                                                                                                                                                                                                            |
| Overview       | Develop / Extract (from a library) a detailed road network graph                                                                                                                                                        |
| Actors         | System                                                                                                                                                                                                                  |
| Pre-Condition  | Appropriate data library is active.                                                                                                                                                                                     |
| Flow           | <p>Main Flow</p> <ol style="list-style-type: none"> <li>1. The system retrieves road network data.</li> <li>2. The data is structured into appropriate structure and stored for future route calculations.</li> </ol>   |
|                | <p>Alternate Flow</p> <ol style="list-style-type: none"> <li>1a. If the data fetch fails, the system retries fetching or uses a cached version (if available). After certain retries, the admin is notified.</li> </ol> |
| Post-Condition | A structured road network graph is ready for routing operations.                                                                                                                                                        |

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| UseCase Number | UC-003                                                                                                                                                                                                                                                                                                    |
| Use Case Name  | Employee Pick-up / Drop-off Location Preference                                                                                                                                                                                                                                                           |
| Overview       | Option for the employees to set their preferred pick-up / drop-off location                                                                                                                                                                                                                               |
| Actors         | Employee, System                                                                                                                                                                                                                                                                                          |
| Pre-Condition  | The employee is registered in the system.                                                                                                                                                                                                                                                                 |
| Flow           | <p>Main Flow</p> <ol style="list-style-type: none"> <li>1. The employee logs into the system and navigates to the preferences section.</li> <li>2. The employee inputs a preferred pickup and drop-off location.</li> <li>3. The input is verified by the system against Google Geocoding API.</li> </ol> |

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|                | <p>Alternate Flow</p> <p>3a. If the inputted address could not be geocoded, the employee would be prompted to enter a reworded address string; until a geocode is obtained.</p> <p>*a. Any errors, at any stage, should raise appropriate error messages and no backend change should go through.</p> |
| Post-Condition | Employee preference(s) are saved for future routing.                                                                                                                                                                                                                                                  |

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| UseCase Number | UC-004                                                                                                                                                                                                                                                                         |
| Use Case Name  | Routing (Many-to-One)                                                                                                                                                                                                                                                          |
| Overview       | Choosing and implementing an optimal algorithm for routing (a singular vehicle) through the extracted longitudes and latitudes                                                                                                                                                 |
| Actors         | System                                                                                                                                                                                                                                                                         |
| Pre-Condition  | Road network graphs and geocoded employee locations are available.                                                                                                                                                                                                             |
| Flow           | <p>Main Flow</p> <ol style="list-style-type: none"> <li>1. The system retrieves the locations of employees to be picked up.</li> <li>2. The routing algorithm calculates the optimal path using Google Routes API..</li> <li>3. Route path is generated and stored.</li> </ol> |
|                | <p>Alternate Flow</p> <p>NA</p>                                                                                                                                                                                                                                                |
| Post-Condition | The optimal Many-to-One route is generated and stored.                                                                                                                                                                                                                         |

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| UseCase Number | UC-005                                                                     |
| Use Case Name  | Visualise Route                                                            |
| Overview       | Appropriately visualise the routes with detailing for driver and employees |
| Actors         | System, Driver, Employee                                                   |
| Pre-Condition  | Routing is complete.                                                       |

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| Flow           | Main Flow <ol style="list-style-type: none"> <li>1. The system fetches the generated route.</li> <li>2. The route is displayed on an interactive map interface.</li> <li>3. Employees and drivers can view the planned route.</li> </ol> |
|                | Alternate Flow <ol style="list-style-type: none"> <li>2a. If the map service fails, textual directions are provided.</li> </ol>                                                                                                          |
| Post-Condition | Routes are successfully displayed for all relevant users.                                                                                                                                                                                |

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| UseCase Number | UC-006                                                                                                                                                                                                                             |
| Use Case Name  | Adding Constraints                                                                                                                                                                                                                 |
| Overview       | Fine tune the algorithm to work with predefined constraints like vehicle capacity, maximum delay and wait time                                                                                                                     |
| Actors         | System, System Administrator                                                                                                                                                                                                       |
| Pre-Condition  | Routing system is operational.                                                                                                                                                                                                     |
| Flow           | Main Flow <ol style="list-style-type: none"> <li>1. The administrator defines constraints such as vehicle capacity and wait times.</li> <li>2. The routing algorithm incorporates these constraints for future routing.</li> </ol> |
|                | Alternate Flow <ol style="list-style-type: none"> <li>*a. If constraints make routing impossible, the admin is notified.</li> </ol>                                                                                                |
| Post-Condition | Constraints are successfully applied to routing.                                                                                                                                                                                   |

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| UseCase Number | UC-007                                                                             |
| Use Case Name  | Ride History and Reports                                                           |
| Overview       | Employees and admins can view past rides, including routes taken and trip details. |
| Actors         | Employee, System Administrator                                                     |
| Pre-Condition  | The system has rides data stored                                                   |
| Flow           | Main Flow                                                                          |

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|                | <ol style="list-style-type: none"> <li>1. Users navigate to the ride history section.</li> <li>2. The system fetches and displays past ride details.</li> <li>3. Users can filter by date, location, or driver.</li> </ol> |
|                | Alternate Flow<br>1a. If no records exist, an appropriate message is displayed.                                                                                                                                            |
| Post-Condition | Ride history is displayed successfully.                                                                                                                                                                                    |

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| UseCase Number | UC-008                                                                                                                                                                                                                                                 |
| Use Case Name  | Many-to-Many Routing                                                                                                                                                                                                                                   |
| Overview       | Multiple vehicle routing                                                                                                                                                                                                                               |
| Actors         | System                                                                                                                                                                                                                                                 |
| Pre-Condition  | Multiple vehicles and employee locations are available.                                                                                                                                                                                                |
| Flow           | Main Flow <ol style="list-style-type: none"> <li>1. The system groups employees based on proximity and drop-off location.</li> <li>2. Routes are assigned to multiple vehicles.</li> <li>3. The system ensures optimal vehicle utilization.</li> </ol> |
|                | Alternate Flow<br>*a. In case of any error, appropriate errors are displayed. The user may re-run the functionality.                                                                                                                                   |
| Post-Condition | Many-to-Many routes are generated successfully.                                                                                                                                                                                                        |

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| UseCase Number | UC-009                                                                                                            |
| Use Case Name  | User Interface                                                                                                    |
| Overview       | Develop interactive and user-friendly web-based applications for driver, employees and system administrator       |
| Actors         | System, Users (Employees, Drivers, Admins)                                                                        |
| Pre-Condition  | Backend functionality is complete.                                                                                |
| Flow           | Main Flow <ol style="list-style-type: none"> <li>1. UI is designed and implemented for each user role.</li> </ol> |

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|                | <ol style="list-style-type: none"> <li>Employees and drivers can access their respective routes, stops, passengers and vehicle details.</li> <li>Admins can view the status of all vehicles, drivers, vehicles and routes.</li> </ol> |
|                | Alternate Flow<br>NA                                                                                                                                                                                                                  |
| Post-Condition | Users can interact seamlessly with the system.                                                                                                                                                                                        |

## Use Case Diagram

